

HEALTHPLUS CARE HEALTHCARE ANALYTICS PROJECT

SQL DATA ANALYSIS REPORT

Prepared By:

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Technology Used:

MySQL

Dataset:

HealthPlus Care Dataset

EXECUTIVE SUMMARY:

This report presents a comprehensive SQL-based analysis of the HealthPlus Care healthcare management dataset. The analysis covers consultations, telemedicine services, chronic-care programs, package subscriptions, prescriptions, laboratory tests, insurance claims, billing, payments, and member feedback.

The objective of this project is to identify operational trends, financial performance, healthcare utilization patterns, and patient satisfaction metrics through data-driven analysis. The findings and insights generated from this project support better decision-making and healthcare service optimization.

KPI SUMMARY DASHBOARD

KPI	VALUES
TOTAL MEMBERS	2500
TOTAL CONSULTATIONS	6000
TOTAL REVENUE	₹15,953,638.11
TOTAL PAYMENT COLLECTION	₹15,953,638.11
COLLECTION GAP	₹0.00
AVERAGE RATING	3.80
TOP CLINIC	HealthPlus Diagnostic Clinic (1243 Consultations)
TOP SPECIALIST	Charu Rao (391 Consultations)
MOST PRESCRIBED MEDICINE	Paracetamol 650mg (554)
TOP INSURANCE PROVIDER	Niva Bupa Health Insurance (₹2,133,157.01)

BUSINESS RECOMMENDATIONS

1. Increase specialist availability in high-volume clinics.
2. Promote telemedicine services in low-adoption regions.
3. Strengthen chronic-care follow-up processes.
4. Improve billing collection efficiency.
5. Monitor high-cost laboratory tests.
6. Improve low- rated service areas.
7. Focus on highly subscribed health packages.
8. Strengthen insurance-provider partnerships.

Step 1: Category-wise Analysis Queries

Question 1

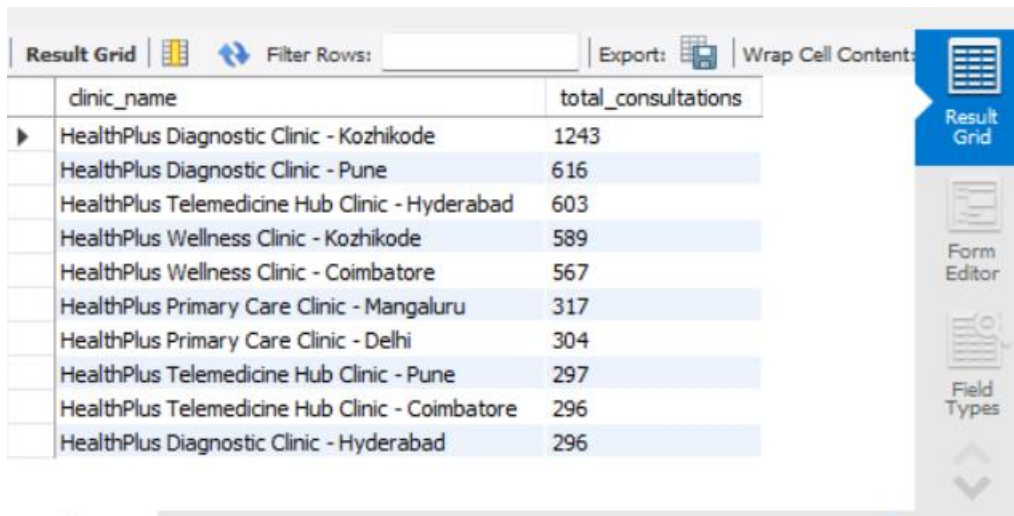
Business Question:

Which clinics have the highest consultation volume?

SQL Query:

```
SELECT
    c.clinic_name,
    COUNT(co.consultation_id) AS total_consultations
FROM Clinics c
JOIN Consultations co
ON c.clinic_id = co.clinic_id
GROUP BY c.clinic_name
ORDER BY total_consultations DESC
LIMIT 10;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab selected. The grid displays the results of the SQL query, listing the top 10 clinics by total consultations in descending order. The columns are 'clinic_name' and 'total_consultations'. The interface also includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Contents' checkbox. On the right side, there are buttons for 'Result Grid', 'Form Editor', and 'Field Types'.

clinic_name	total_consultations
HealthPlus Diagnostic Clinic - Kozhikode	1243
HealthPlus Diagnostic Clinic - Pune	616
HealthPlus Telemedicine Hub Clinic - Hyderabad	603
HealthPlus Wellness Clinic - Kozhikode	589
HealthPlus Wellness Clinic - Coimbatore	567
HealthPlus Primary Care Clinic - Mangaluru	317
HealthPlus Primary Care Clinic - Delhi	304
HealthPlus Telemedicine Hub Clinic - Pune	297
HealthPlus Telemedicine Hub Clinic - Coimbatore	296
HealthPlus Diagnostic Clinic - Hyderabad	296

Business Insight:

This analysis identifies the clinics handling the highest consultation workload. High-volume clinics indicate stronger member engagement and healthcare service demand.

Question 2

Business Question:

Which specialists have the highest consultation workload?

SQL Query:

```
SELECT
    s.specialist_id,
    CONCAT(s.first_name, ' ', s.last_name) AS specialist_name,
    COUNT(c.consultation_id) AS consultation_count
FROM Specialists s
JOIN Consultations c
ON s.specialist_id = c.specialist_id
GROUP BY s.specialist_id, specialist_name
ORDER BY consultation_count DESC;
```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
specialist_id	specialist_name	consultation_count	
SP0004	Charu Rao	391	
SP0093	Sameer Malhotra	70	
SP0080	Rajat Saxena	66	
SP0133	Sanya Ahluwalia	65	
SP0112	Nikhil Menon	62	
SP0194	Sanya Bhatia	59	
SP0123	Anjali Choudhary	55	
SP0088	Meghna Choudhary	53	
SP0086	Diya Bhatia	52	
SP0155	Preeti Kapoor	48	
SP0173	Priyal Tiwari	48	
SP0118	Mansi Desai	48	
SP0148	Arvind Bhatia	46	
SP0220	Gaurav Kulkarni	46	
SP0055	Kirti Chawla	46	
SP0116	Sneha Bhatia	45	
SP0085	Ishaan Tiwari	45	
SP0074	Mansi Mishra	44	
SP0082	Ritu Choudhary	44	
SP0172	Mansi Gupta	43	
SP0021	Puneet Bhatia	43	
SP0217	Rajat Malhotra	43	
SP0109	Falguni Mishra	42	
SP0103	Karan Rao	40	
SP0211	Kabir Iyer	38	
SP0113	Riya Desai	38	
SP0010	Garima Bhatia	37	
SP0058	Riya Malhotra	37	
SP0174	Tanvi Mehta	37	
SP0190	Sameer Gupta	37	
SP0140	Amit Malhotra	36	
SP0192	Gaurav Menon	35	
SP0072	Sameer Joshi	35	

Business Insight:

Specialists with the highest consultation counts contribute significantly to healthcare delivery and may require workload balancing and resource support.

Question 3

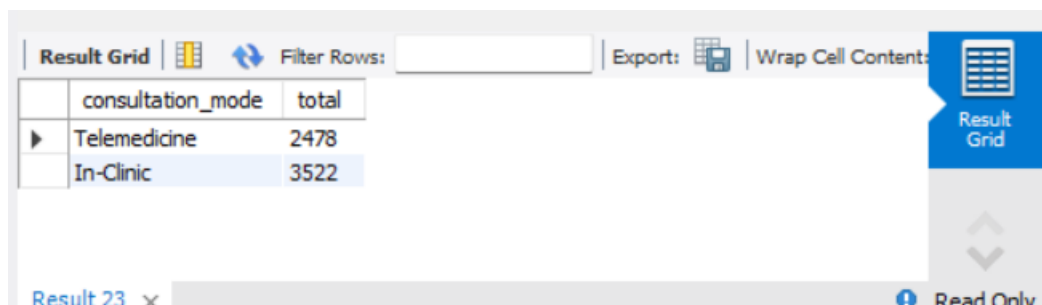
Business Question:

How are consultations distributed across different consultation modes?

SQL Query:

```
SELECT  
consultation_mode,  
COUNT(*) AS total  
FROM Consultations  
GROUP BY consultation_mode;
```

Output:



The screenshot shows a 'Result Grid' interface with a table containing two rows of data. The table has columns 'consultation_mode' and 'total'. The first row is 'Telemedicine' with a total of 2478, and the second row is 'In-Clinic' with a total of 3522. The interface includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Contents' button. A 'Result Grid' button is also visible on the right side of the table.

consultation_mode	total
Telemedicine	2478
In-Clinic	3522

Business Insight:

This analysis helps understand member preference between consultation modes and supports service planning decisions.

Step 2: Telemedicine Analysis

Question 4

Business Question: What is the average duration of telemedicine sessions?

SQL Query:

```

SELECT
  AVG(
    TIMESTAMPDIFF(
      MINUTE,
      session_start_time,
      session_end_time
    ) AS avg_duration_minutes
  ) AS avg_duration_minutes
FROM Telemedicine_Sessions;

```

Output:

The screenshot shows a SQL query result grid. The header row contains the column name 'avg_duration_minutes'. The first data row shows the value '21.4273'. The interface includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Contents' option. A 'Result Grid' button is visible on the right side. The bottom of the window shows 'Result 25' and a 'Read Only' status.

avg_duration_minutes
21.4273

Business Insight:

Average session duration helps evaluate virtual consultation effectiveness and operational efficiency.

Question 5

Business Question:

Which telemedicine platforms are used most frequently?

SQL Query:

```

SELECT
  platform,
  COUNT(*) total_sessions
FROM Telemedicine_Sessions
GROUP BY platform
ORDER BY total_sessions DESC;

```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
platform	total_sessions		
App	792		
Phone Call	731		
Web	677		

Result 26 x Read Only

Business Insight:

Understanding platform usage helps management identify the most preferred virtual healthcare channels.

Step 3: Chronic Care Analysis

Question 6

Business Question:

Which chronic conditions have the highest member enrollment?

SQL Query:

```
SELECT
condition_name,
COUNT(*) total_members
FROM Chronic_Care_Programs
GROUP BY condition_name
ORDER BY total_members DESC;
```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
condition_name	total_members		
Hypertension	120		
Type 2 Diabetes	108		
Asthma	107		
Obesity Management	102		
Arthritis	100		
Thyroid Disorder	96		
Chronic Kidney Disease	92		
Chronic Back Pain	90		
Cardiac Risk Management	85		

Result 28 x Read Only

Business Insight:

Frequently occurring conditions indicate major healthcare concerns among members and help prioritize chronic care initiatives.

Question 7

Business Question:

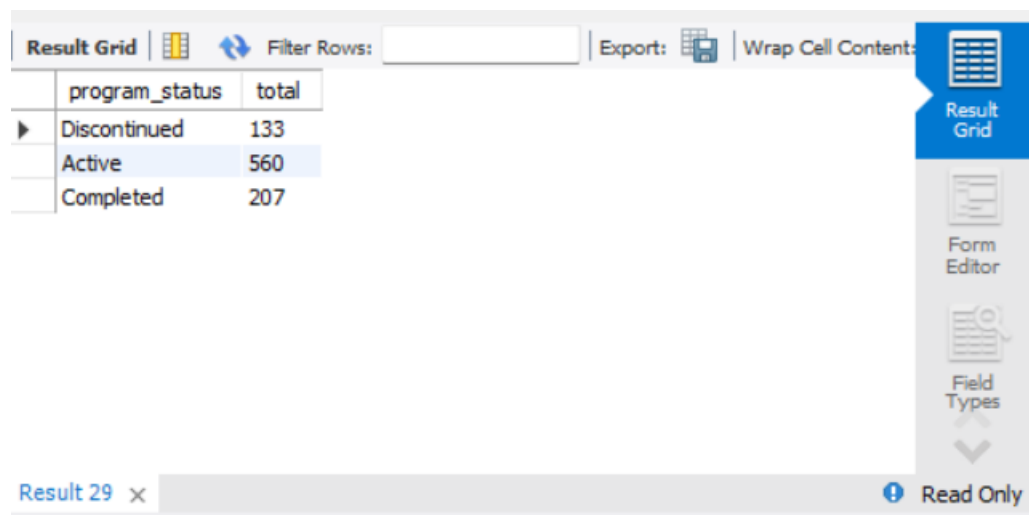
What is the distribution of chronic care program statuses?

SQL Query:

```
-- Active vs Inactive Programs

SELECT
program_status,
COUNT(*) total
FROM Chronic_Care_Programs
GROUP BY program_status;
```

Output:



The screenshot shows a SQL query result grid with the following data:

program_status	total
Discontinued	133
Active	560
Completed	207

The interface includes a toolbar with options like 'Filter Rows', 'Export', and 'Wrap Cell Content'. A sidebar on the right contains icons for 'Result Grid', 'Form Editor', and 'Field Types'. The bottom status bar indicates 'Result 29' and 'Read Only'.

Business Insight:

Program status analysis provides visibility into active, inactive, completed, or pending chronic care programs.

Step 4: Package Analysis

Question 8

Business Question:

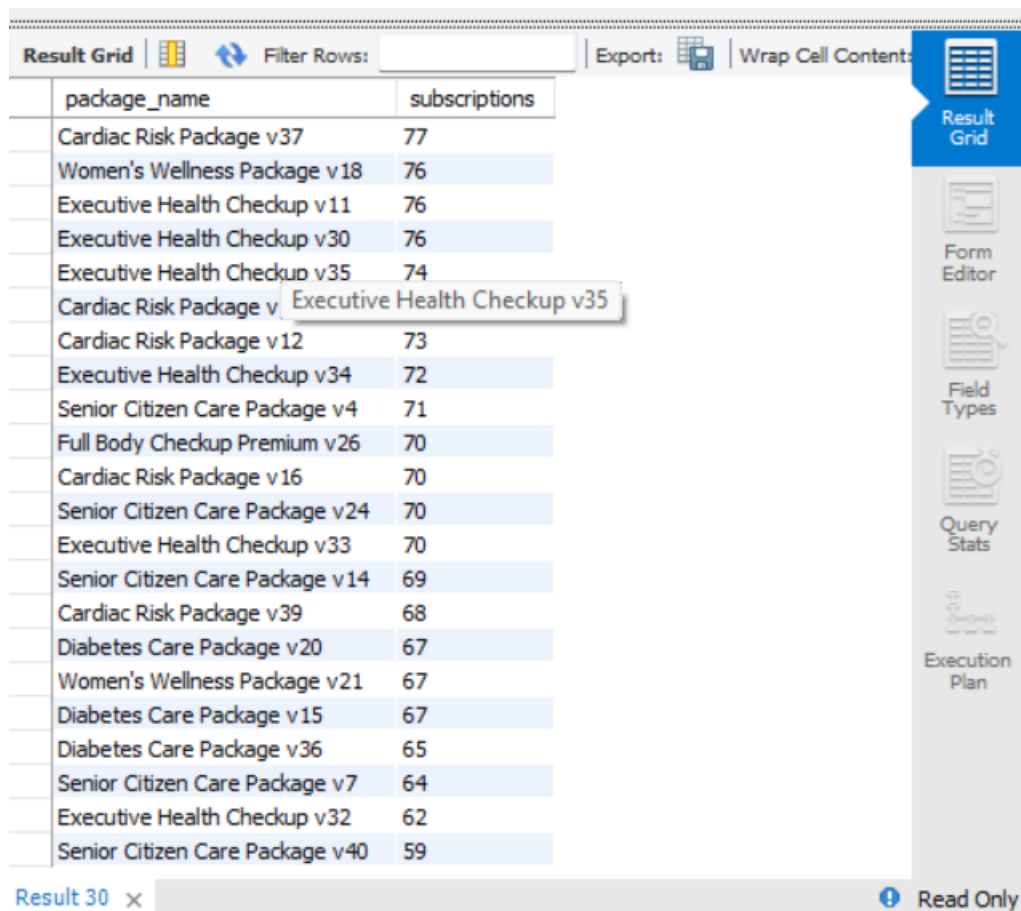
Which health packages have the highest subscription volume?

SQL Query:

```
-- Most Subscribed Packages

SELECT
  hp.package_name,
  COUNT(ps.subscription_id) subscriptions
FROM Health_Packages hp
JOIN Package_Subscriptions ps
ON hp.package_id = ps.package_id
GROUP BY hp.package_name
ORDER BY subscriptions DESC;
```

Output:



package_name	subscriptions
Cardiac Risk Package v37	77
Women's Wellness Package v18	76
Executive Health Checkup v11	76
Executive Health Checkup v30	76
Executive Health Checkup v35	74
Cardiac Risk Package v12	73
Executive Health Checkup v34	72
Senior Citizen Care Package v4	71
Full Body Checkup Premium v26	70
Cardiac Risk Package v16	70
Senior Citizen Care Package v24	70
Executive Health Checkup v33	70
Senior Citizen Care Package v14	69
Cardiac Risk Package v39	68
Diabetes Care Package v20	67
Women's Wellness Package v21	67
Diabetes Care Package v15	67
Diabetes Care Package v36	65
Senior Citizen Care Package v7	64
Executive Health Checkup v32	62
Senior Citizen Care Package v40	59

Business Insight:

Popular healthcare packages reveal member preferences and assist management in optimizing package offerings.

Step 5: Corporate Analysis

Question 9

Business Question:

Which corporate clients contribute the highest member enrollment?

SQL Query:

```
-- Top Corporate Clients

SELECT
c.company_name,
COUNT(cm.member_id) total_members
FROM Corporates c
JOIN Corporate_Members cm
ON c.corporate_id=cm.corporate_id
GROUP BY c.company_name
ORDER BY total_members DESC;
```

Output:

company_name	total_members
SkyIndustries Pvt Ltd	41
NovaSolutions Pvt Ltd	39
ApexSolutions Pvt Ltd	36
TechSolutions Pvt Ltd	35
DeltaSystems Pvt Ltd	30
TechSystems Pvt Ltd	28
NovaIndustries Pvt Ltd	27
SkySoft Pvt Ltd	26
DeltaSolutions Pvt Ltd	25
ApexLogistics Pvt Ltd	23
ZenIndustries Pvt Ltd	22
DeltaSoft Pvt Ltd	21
ApexCorp Pvt Ltd	21
SkyLogistics Pvt Ltd	19
ZenSolutions Pvt Ltd	18
GlobalSoft Pvt Ltd	18
SkySolutions Pvt Ltd	12
PrimeSolutions Pvt Ltd	12
ApexSystems Pvt Ltd	12
GlobalLogistics Pvt Ltd	11
TechCorp Pvt Ltd	11
TechLogistics Pvt Ltd	11
NovaCorp Pvt Ltd	10
NovaSoft Pvt Ltd	10
PrimeSoft Pvt Ltd	10
ZenCorp Pvt Ltd	9
ZenSoft Pvt Ltd	9
PrimeIndustries Pvt Ltd	9
ApexSoft Pvt Ltd	9
DeltaCorp Pvt Ltd	8
DeltaLogistics Pvt Ltd	8
GlobalSolutions Pvt Ltd	8

Business Insight:

Corporates with the highest enrollment contribute significantly to business growth and healthcare service utilization.

Step 6: Prescription Analysis

Question 10

Business Question:

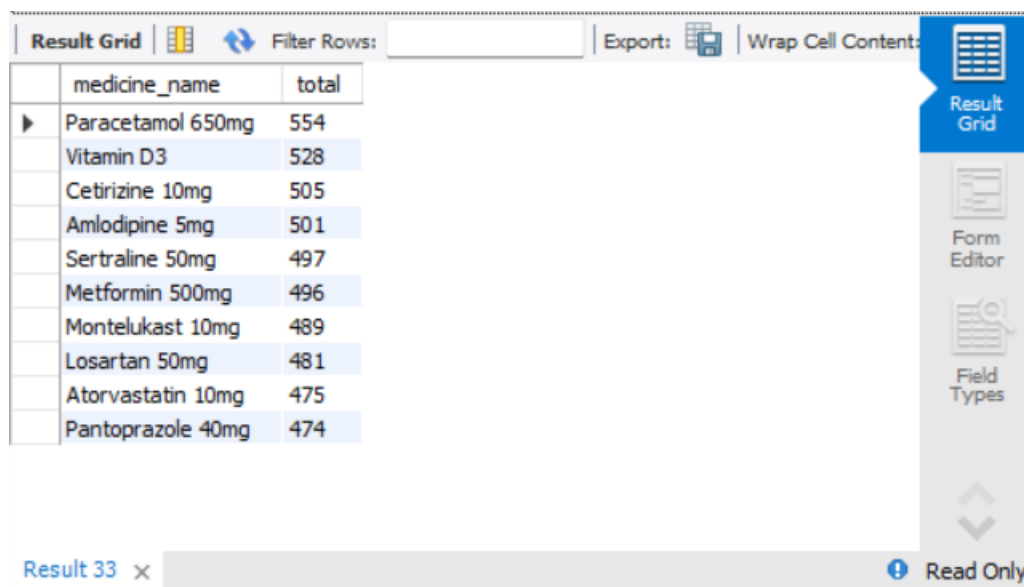
Which medicines are prescribed most frequently?

SQL Query:

```
-- Most Prescribed Medicines

SELECT
medicine_name,
COUNT(*) total
FROM Prescriptions
GROUP BY medicine_name
ORDER BY total DESC
LIMIT 10;
```

Output:



	medicine_name	total
▶	Paracetamol 650mg	554
	Vitamin D3	528
	Cetirizine 10mg	505
	Amlodipine 5mg	501
	Sertraline 50mg	497
	Metformin 500mg	496
	Montelukast 10mg	489
	Losartan 50mg	481
	Atorvastatin 10mg	475
	Pantoprazole 40mg	474

Business Insight:

Frequently prescribed medicines help identify common treatment patterns and healthcare needs among members.

Step 7: Lab Test Analysis

Question 11

Business Question:

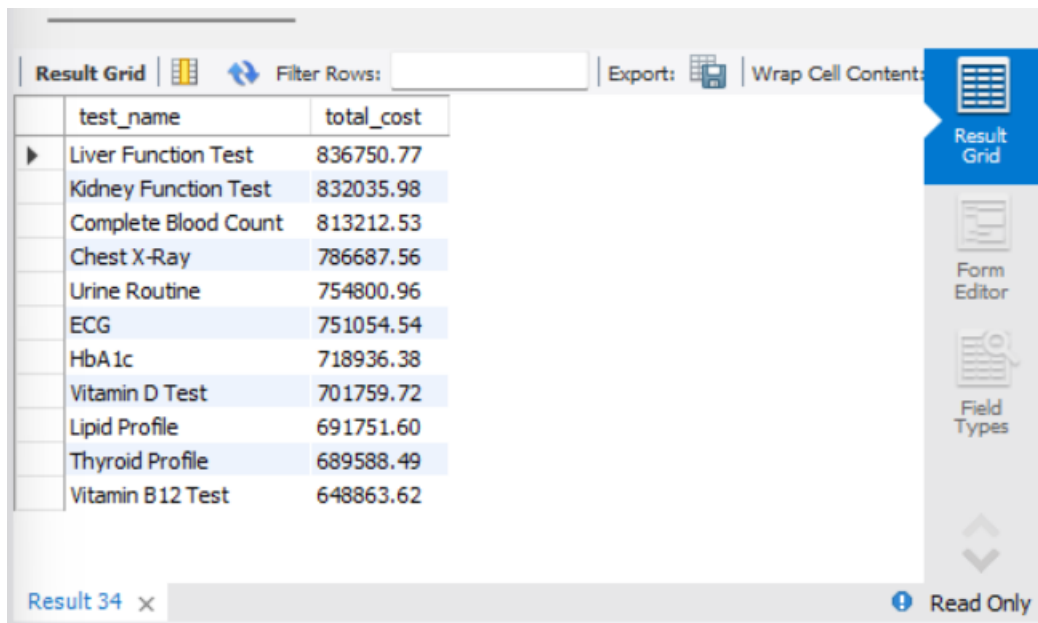
Which laboratory tests generate the highest total diagnostic cost?

SQL Query:

```
-- Highest Cost Tests

SELECT
test_name,
SUM(test_cost) total_cost
FROM Lab_Tests
GROUP BY test_name
ORDER BY total_cost DESC;
```

Output:



	test_name	total_cost
▶	Liver Function Test	836750.77
	Kidney Function Test	832035.98
	Complete Blood Count	813212.53
	Chest X-Ray	786687.56
	Urine Routine	754800.96
	ECG	751054.54
	HbA1c	718936.38
	Vitamin D Test	701759.72
	Lipid Profile	691751.60
	Thyroid Profile	689588.49
	Vitamin B12 Test	648863.62

Business Insight:

High-cost diagnostic tests contribute significantly to healthcare spending and may require cost monitoring.

Step 8: Claims Analysis

Question 12

Business Question:

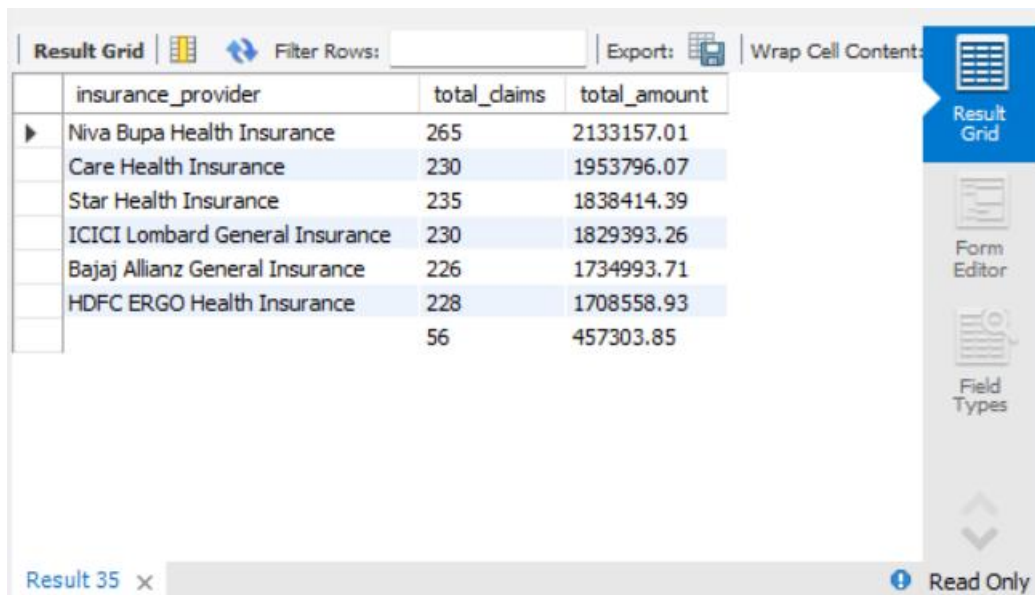
Which insurance providers handle the highest claim volume and claim amount?

SQL Query:

```
-- Insurance Provider Performance

SELECT
insurance_provider,
COUNT(*) total_claims,
SUM(claim_amount) total_amount
FROM Claims
GROUP BY insurance_provider
ORDER BY total_amount DESC;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab selected. The grid displays the results of the SQL query, ordered by total claim amount in descending order. The columns are 'insurance_provider', 'total_claims', and 'total_amount'. The data is as follows:

insurance_provider	total_claims	total_amount
Niva Bupa Health Insurance	265	2133157.01
Care Health Insurance	230	1953796.07
Star Health Insurance	235	1838414.39
ICICI Lombard General Insurance	230	1829393.26
Bajaj Allianz General Insurance	226	1734993.71
HDFC ERGO Health Insurance	228	1708558.93
	56	457303.85

At the bottom of the grid, it says 'Result 35' and 'Read Only'.

Business Insight:

This analysis identifies major insurance partners and helps evaluate claim-processing performance.

Step 9: Revenue Analysis

Question 13

Business Question:

What is the total billed revenue generated by HealthPlus Care?

SQL Query:

```
-- Total Revenue

SELECT
SUM(total_amount) total_billing
FROM Billing;
```

Output:

Result Grid		Filter Rows:
	total_billing	
▶	15953638.11	

Result 36 x

Business Insight:

Total billing represents the overall revenue generated from healthcare services during the reporting period.

Question 14

Business Question:

What is the total payment amount collected from members?

SQL Query:

```
-- Total Payment Collection

SELECT
SUM(payment_amount) total_payment
FROM Payments;
```

Output:

Result Grid		Filter Rows:
	total_payment	
▶	15953638.11	

Result 37 x

Business Insight:

Payment collection indicates how much of the billed revenue has been successfully recovered.

Question 15

Business Question:

What is the collection gap between billed amounts and received payments?

SQL Query:

```
-- Collection Gap

SELECT
  (SELECT SUM(total_amount)FROM Billing)
-
  (SELECT SUM(payment_amount)FROM Payments)
AS collection_gap;
```

Output:

```
-- Collection Gap

SELECT
  (SELECT SUM(total_amount)FROM Billing)
-
  (SELECT SUM(payment_amount)FROM Payments)
AS collection_gap;
```

Business Insight:

A higher collection gap may indicate delayed payments, pending collections, or financial inefficiencies that require management attention.

Step 10: Feedback Analysis

Question 16

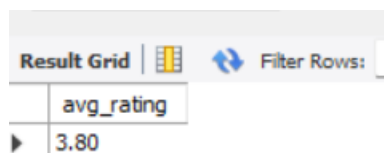
Business Question:

What is the average feedback rating provided by members?

SQL Query:

```
-- Average Rating  
  
SELECT  
  ROUND(AVG(rating),2)  
  AS avg_rating  
FROM Feedback;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. The grid has one column labeled 'avg_rating' and one row with the value '3.80'. There is a 'Filter Rows' button and a small icon to the right of the column header.

avg_rating
3.80

Result 39 x

Business Insight:

Average ratings provide an overall measure of member satisfaction and healthcare service quality.

Question 17

Business Question:

Which specialists receive the highest average ratings from members?

SQL Query:

```
-- Best Specialists

SELECT
s.specialist_id,
CONCAT(s.first_name, ' ', s.last_name) specialist_name,
ROUND(AVG(f.rating), 2) avg_rating
FROM Specialists s
JOIN Consultations c
ON s.specialist_id=c.specialist_id
JOIN Feedback f
ON c.consultation_id=f.consultation_id
GROUP BY s.specialist_id, specialist_name
ORDER BY avg_rating DESC;
```

Output:

specialist_id	specialist_name	avg_rating
SP0175	Priyal Chawla	5.00
SP0001	Nidhi Mishra	5.00
SP0128	Rahul Bhatia	5.00
SP0105	Puneet Iyer	4.80
SP0102	Aditi Chopra	4.67
SP0051	Diya Mishra	4.58
SP0200	Nikhil Verma	4.57
SP0164	Shaurya Menon	4.56
SP0136	Ankit Malhotra	4.50
SP0130	Nisha Shah	4.50
SP0179	Puneet Menon	4.50
SP0196	Mansi Iyer	4.50
SP0027	Vivek Choudhary	4.50
SP0212	Isha Iyer	4.50
SP0209	Charu Menon	4.50
SP0123	Anjali Choudhary	4.44
SP0216	Arvind Rao	4.40
SP0201	Charu Agarwal	4.36
SP0135	Priyal Ahluwalia	4.33
SP0013	Anjali Kapoor	4.33
SP0178	Garima Nair	4.29
SP0199	Priyal Reddy	4.29
SP0037	Siddharth Bansal	4.29
SP0006	Preeti Mishra	4.25
SP0012	Tushar Chawla	4.25
SP0148	Arvind Bhatia	4.25
SP0020	Priyal Bhatia	4.25
SP0115	Ojasvi Choudh...	4.22
SP0069	Diya Rao	4.20
SP0161	Sanya Tiwari	4.18
SP0042	Esha Agarwal	4.18
SP0143	Esha Iyer	4.17
SP0112	Nikhil Menon	4.15
SP0106	Esha Iyer	4.14

Business Insight:

Highly rated specialists demonstrate strong service quality and positive patient experiences.

Step 11: Window Function Analysis

Question 18

Business Question:

What is the sequence of consultations for each member?

SQL Query:

```
-- ROW_NUMBER()

SELECT
member_id,
consultation_date,
ROW_NUMBER()
OVER(PARTITION BY member_id ORDER BY consultation_date) AS visit_no
FROM Consultations;
```

Output:


```
-- RANK()

SELECT
specialist_id,
COUNT(*) total_consultations,
RANK()
OVER(ORDER BY COUNT(*) DESC) ranking
FROM Consultations
GROUP BY specialist_id;
```

Output:

The screenshot shows a database interface with a 'Result Grid' tab selected. The grid displays the results of the SQL query, listing specialists by their ID, total consultations, and a calculated ranking. The ranking is based on the total consultations in descending order. The interface includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Contents' option. On the right side, there is a vertical toolbar with icons for 'Result Grid', 'Form Editor', 'Field Types', and 'Query Stats'. At the bottom, it indicates 'Result 42' and a 'Read Only' status.

specialist_id	total_consultations	ranking
SP0004	391	1
SP0093	70	2
SP0080	66	3
SP0133	65	4
SP0112	62	5
SP0194	59	6
SP0123	55	7
SP0088	53	8
SP0086	52	9
SP0118	48	10
SP0155	48	10
SP0173	48	10
SP0055	46	13
SP0148	46	13
SP0220	46	13

Business Insight:

Ranking specialists helps identify top-performing healthcare providers and workload distribution.

Question 20

Business Question:

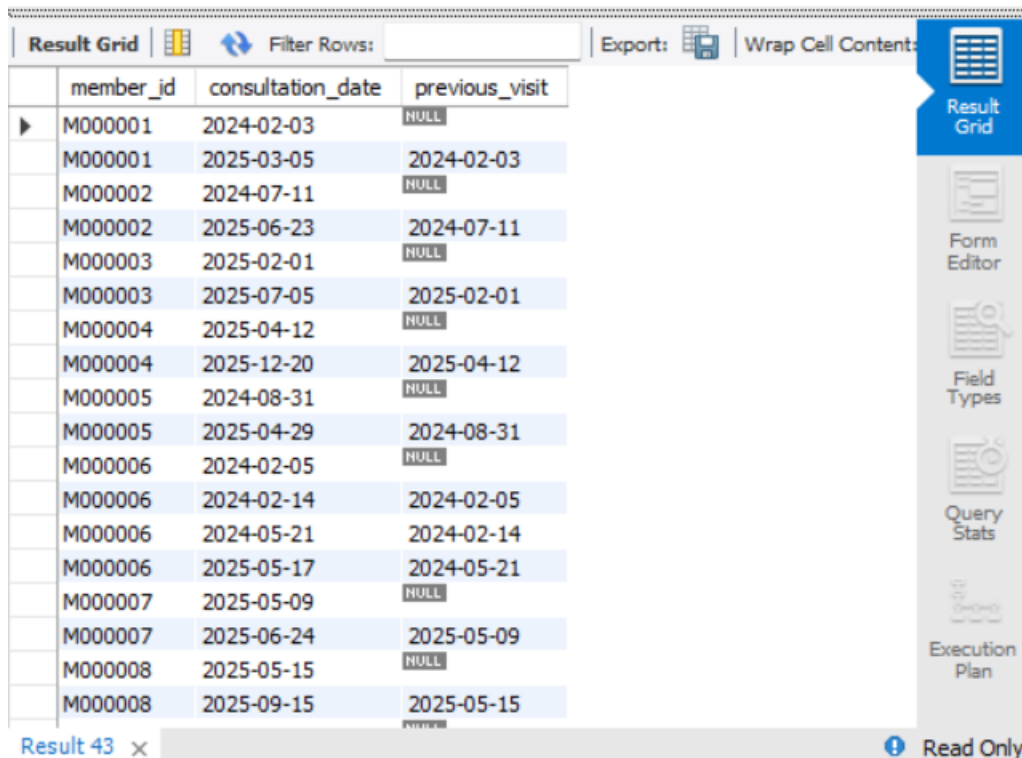
What was the previous consultation date for each member?

SQL Query:

```
-- LAG()

SELECT
member_id,
consultation_date,
LAG(consultation_date)
OVER(PARTITION BY member_id ORDER BY consultation_date) previous_visit
FROM Consultations;
```

Output:



The screenshot shows a database query result grid with the following columns: member_id, consultation_date, and previous_visit. The grid contains 16 rows of data. The 'previous_visit' column shows the next consultation date for each member, with NULL values indicating no further visits recorded. The interface includes a 'Result Grid' tab, a 'Filter Rows' section, and a 'Read Only' status at the bottom right.

	member_id	consultation_date	previous_visit
▶	M000001	2024-02-03	NULL
	M000001	2025-03-05	2024-02-03
	M000002	2024-07-11	NULL
	M000002	2025-06-23	2024-07-11
	M000003	2025-02-01	NULL
	M000003	2025-07-05	2025-02-01
	M000004	2025-04-12	NULL
	M000004	2025-12-20	2025-04-12
	M000005	2024-08-31	NULL
	M000005	2025-04-29	2024-08-31
	M000006	2024-02-05	NULL
	M000006	2024-02-14	2024-02-05
	M000006	2024-05-21	2024-02-14
	M000006	2025-05-17	2024-05-21
	M000007	2025-05-09	NULL
	M000007	2025-06-24	2025-05-09
	M000008	2025-05-15	NULL
	M000008	2025-09-15	2025-05-15

Business Insight:

LAG analysis helps understand member revisit patterns and healthcare engagement frequency.

Question 21

Business Question:

What is the next consultation date for each member?

SQL Query:

```
-- LEAD()

SELECT
  member_id,
  consultation_date,
  LEAD(consultation_date)
  OVER(PARTITION BY member_id ORDER BY consultation_date) next_visit
FROM Consultations;
```

Output:

Result Grid			
Filter Rows:			
Export: Wrap Cell Contents			
	member_id	consultation_date	next_visit
▶	M000001	2024-02-03	2025-03-05
	M000001	2025-03-05	NULL
	M000002	2024-07-11	2025-06-23
	M000002	2025-06-23	NULL
	M000003	2025-02-01	2025-07-05
	M000003	2025-07-05	NULL
	M000004	2025-04-12	2025-12-20
	M000004	2025-12-20	NULL
	M000005	2024-08-31	2025-04-29
	M000005	2025-04-29	NULL
	M000006	2024-02-05	2024-02-14
	M000006	2024-02-14	2024-05-21
	M000006	2024-05-21	2025-05-17
	M000006	2025-05-17	NULL
	M000007	2025-05-09	2025-06-24
	M000007	2025-06-24	NULL
	M000008	2025-05-15	2025-09-15
	M000008	2025-09-15	NULL
	M000009	2025-05-31	NULL
	M000010	2025-10-06	NULL
	M000011	2024-12-07	2025-09-17

Business Insight:

LEAD analysis helps forecast upcoming healthcare interactions and supports follow-up planning.

CONCLUSION:

PROJECT CONCLUSION

The HealthPlus Care project successfully analyzed healthcare operations using SQL-based techniques. The project included data profiling, validation, analytical querying, and business insight generation across consultations, telemedicine services, chronic-care programs, prescriptions, claims, billing, payments, and member feedback.

The analysis helped identify healthcare utilization patterns, financial performance indicators, and member satisfaction trends. The generated insights and recommendations can support better operational planning, resource allocation, and service-quality improvements.

Overall, the project demonstrates the effective use of SQL for transforming healthcare data into actionable business insights that support data-driven decision-making.